

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India 09/2021 – 01/2025 (3.5 Years)

- Supported the **Linux yocto build process**, applying necessary patches and customizing the kernel and file system for embedded device firmware.
- Developed and maintained container-based build environments, enhancing CI/CD pipeline efficiency by **15%** through automated Docker image creation.
- Contributed to the development of module requirements, design specifications, and module tests for new embedded software features, ensuring adherence to **MISRA standards**.
- Executed testing at black box, integration, and unit levels for new platform features, identifying and resolving **over 50 defects** prior to release.
- Updated software requirement specifications through collaborative discussions with SRS Engineers, SW Architects, and Senior Developers, improving documentation clarity by **20%**.
- Organized and conducted formal code and document reviews of own work products, implementing feedback to enhance code quality and maintainability.

SKILLS

Embedded Systems – Firmware, Drivers, Middleware, Yocto, Kernel Customization, File System Customization

Programming Languages – C, C++, Python

Build & Toolchain – Linux, Klocwork, CMake, Compiler, Linker

Hardware Interface – GPIO, SPI, I2C, UART, Ethernet, Memory

Documentation & Collaboration – Wiki, Google Sites, Confluence, Jira, Codebeamer

Quality & Standards – ASPICE, ISO15504, MISRA C

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin 02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India 01/2021

PROJECTS

Embedded Linux Kernel & Build System for Automotive TI SoC

- Configured and maintained the **Linux Yocto build environment** for a new Texas Instruments SoC, enabling custom kernel builds and root filesystem generation.
- Developed automation scripts in **Python** to streamline the build process, reducing manual intervention time by **30%** and improving build consistency.
- Integrated the Vision framework onto the target SoC hardware, developing necessary drivers and middleware for device functionality and demonstration.

Automated AI Solution for Platform Code Generation

- Designed and implemented an end-to-end **AI solution** to analyze requirements and generate platform-specific code for customer projects, accelerating development cycles.
- Developed core components using **Python** and integrated with AI models for requirement parsing and code synthesis, achieving a **90%** accuracy rate in initial testing.
- Managed project artifacts and conducted static code analysis using tools like **Klocwork**, ensuring compliance with coding guidelines and quality standards.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)